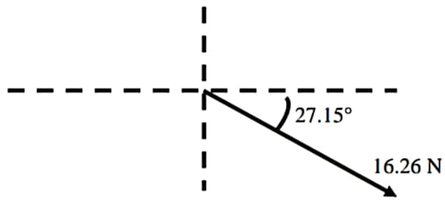
ANSWER SIRI 5 (PSPM SET 3)

NO	ANSWER SCHEME	MA
1 (a)(i)	$\frac{25\ \cancel{g}}{\cancel{cm}^3}\left(\frac{1\ \text{kg}}{1000\ \cancel{g}}\right)\left(\frac{100^3\ \cancel{cm}^3}{1^3\ \text{m}^3}\right) = 25000\ \text{kg m}^{-3}$	J
(a)(ii)	$\frac{8.5\ \cancel{km}}{\cancel{h}}\left(\frac{1000\ \text{m}}{1\ \cancel{km}}\right)\left(\frac{1\ \cancel{h}}{3600\ \text{s}}\right) = 2.36\ \text{m s}^{-1}$	J
(b)	see below	

FORCE	X-Component (N)	Y-Component (N)
P	10 cos 20° = 9.40	-10 sin 20° = -3.42
Q	12	0
R	8 cos 30° = 6.93	8 sin 30° = 4
S	9.4 + 12 – 6.93 = 14.47	-3.42 + 0 - 4 = -7.42

$$|F| = \sqrt{(F_x)^2 + (F_y)^2}$$
$$= \sqrt{(14.47)^2 + (-7.42)^2}$$
$$= 16.26\ \text{N}$$

$$\theta = \tan^{-1}\left(\frac{-7.42}{14.47}\right)$$
$$= -27.15°$$



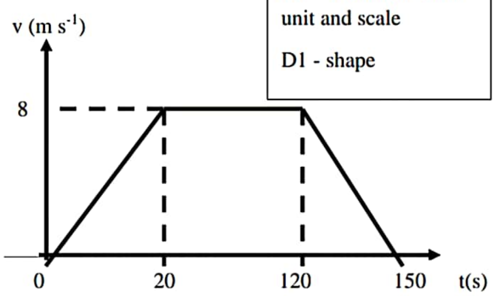
J

TOTAL MARKS

2 (a)(i)

D1 – label axis with unit and scale

D1 - shape



(a)(ii)

$0\ \text{s} - 20\ \text{s} : \frac{v-u}{t} = \frac{8-0}{20} = 0.4\ \text{m s}^{-2}$ J

$20\ \text{s} - 120\ \text{s} : 0\ \text{m s}^{-2}$ J

$120\ \text{s} - 150\ \text{s} : \frac{0-8}{30} = -0.27\ \text{m s}^{-2}$ J

(a)(ii)	$0\text{ s} - 20\text{ s} : \dfrac{v-u}{t} = \dfrac{8-0}{20} = 0.4\text{ m s}^{-2}$ $20\text{ s} - 120\text{ s} : 0\text{ m s}^{-2}$ $120\text{ s} - 150\text{ s} : \dfrac{0-8}{30} = -0.27\text{ m s}^{-2}$ Distance = area under the graph v – t $= \dfrac{1}{2}(150+100)(8) = 1000\text{ m}$
(b)	$v^2 = u^2 + 2as$ $0^2 = (25)^2 + 2(-5)s$ $-625 = -10s$ $s = 62.5\text{ m}$
(c)	$v^2 = u^2 - 2gs$ $0^2 = 3^2 - 2(9.81)s$ $-9 = -19.62s$ $s = 0.46\text{ m}$ $v = u - gt$ $0 = 3 - 9.81t$ $-3 = -9.81t$ $t = 0.31s$
	TOTAL MARKS
3(a)(i)	Impulse = Area under F-t graph $J = \dfrac{1}{2}\times\left[(1.8-0.2)\times10^{-3}\right]\times(5\times10^{3})$ $J = 4\text{ N s}$
(a)(ii)	$J = mv - mu$ $4 = (0.08)(v) - (0.08)(0)$ $v = 50\text{ m s}^{-1}$
(b)	x-comp : $\sum P_i = \sum P_f$ $m_A u_A + m_B u_B = (m_A + m_B)\, v \cos\theta$ $m\,u + m\,(0) = (m + m)\, 8 \cos\theta$ $mu = (2m)\, 8\cos\theta$ $mu = 16m\cos\theta$ $u = 16\cos\theta$ ---- (1)

[redacted]

y-comp : $\sum P_i = \sum P_f$

$$m_A u_A + m_B u_B = (m_A + m_B)\, v \sin\theta$$
$$m\,(0) + m\,(9) = (m + m)\; 8 \sin\theta$$
$$9m = (2m)\; 8 \sin\theta$$
$$9m = 16m \sin\theta$$
$$\sin\theta = 0.563$$
$$\theta = 34.2°$$

From eqn (1)
$$u = 16 \cos\theta$$
$$u = 13.23 \text{ m s}^{-1}$$

TOTAL MARKS

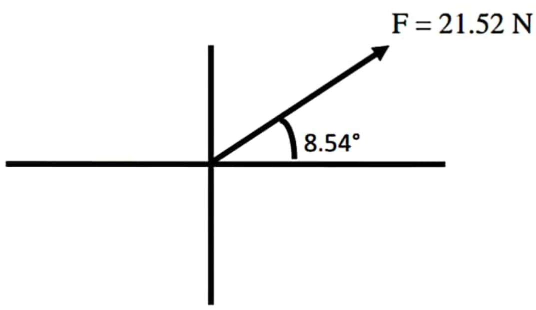
4(a)

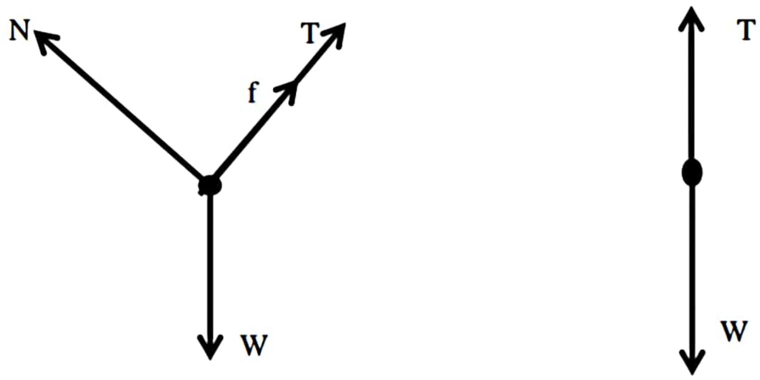
F	x-component (N)	y-component (N)
F_1	-64 cos 45° = -45.25	64 sin 45° = 45.25
F_2	-30 cos 60° = -15	-30 sin 60° = -25.98
F_3	45 cos 30° = 38.97	-45 sin 30° = -22.5
F_4	F_{4x}	F_{4y}
$\sum F$	0	0

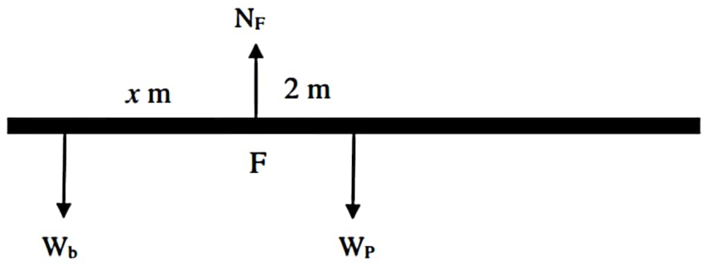
$$\sum F_x = F_{1x} + F_{2x} + F_{3x} + F_{4x}$$
$$0 = -45.25 + (-15) + (38.97) + F_{4x}$$
$$F_{4x} = 21.28\text{N}$$
$$\sum F_y = F_{1y} + F_{2y} + F_{3y} + F_{4y}$$
$$= 45.25 + (-25.98) + (-22.5) + F_{4y}$$
$$F_{4y} = 3.23\text{N}$$

$$F_4 = \sqrt{(F_{4x})^2 + (F_{4y})^2}$$
$$= \sqrt{(21.28)^2 + (3.23)^2}$$
$$= 21.52 \text{ N}$$

$$\theta = \tan^{-1}\left(\frac{3.23}{21.52}\right) = 8.54°$$



(b)(i)	
(b)(ii)	<u>P</u> $\sum F_x = ma$ $W_x - T - f = ma$ $W \sin\theta - T - f = ma$ $3(9.81)\sin 60° - T - 5 = 3a$ $T = 20.49 - 3a$ ------- 1 <u>Q</u> $\sum F_y = ma$ $T - W = ma$ $(20.49 - 3a) - 1(9.81) = 1a$ $a = 2.67 \text{ m s}^{-2}$
	TOTAL MARKS
5 (a)(i)	By using $v = r\omega$ and $\omega = \frac{2\pi}{T}$ $v = \frac{2\pi r}{T}$ $v = \frac{2\pi(5)}{6} = 5.24\, m\, s^{-1}$
(a)(ii)	$F_c = \frac{mv^2}{r}$ $F_c = \frac{\frac{800}{9.81}(5.24)^2}{5}$ $F_c = 447.83\, N$
(a)(iii)	$\sum F_y = 0$ $T_y - W = 0,\ T\cos\theta = 800$ -------------(1)

	$\sum F_x = F_c$ $T\sin\theta = F_c$ $T\sin\theta = 447.83$ ---------(2) $(2)\div(1),\ \dfrac{T\sin\theta}{T\cos\theta} = \dfrac{447.83}{800}$ $\tan\theta = \dfrac{447.83}{800}$ $\theta = 29.24°$	
(b)	$\omega = \dfrac{2\text{ rev}}{\text{s}} = \dfrac{2\text{ rev}}{\text{s}} \times \dfrac{2\pi\text{ rad}}{1\text{ rev}} = 4\pi\text{ rad s}^{-1} = 12.57\text{ rad s}^{-1}$ $a_c = r\omega^2$ $a_c = (1.2)(12.57)^2$ $a_c = 189.6\text{ m s}^{-2}$	
	TOTAL MARKS	
6 (a)(i)	$\omega = \omega_o + \alpha t$ $0 = 10.5 + \alpha(5.2)$ $\alpha = -2.02\text{ rad s}^{-1}$	
6 (a)(ii)	$\theta = \omega_o t + \dfrac{1}{2}\alpha t^2$ $\theta = 10.5(5.2) + \dfrac{1}{2}(-2.02)(5.2)^2$ $\theta = 27.29\text{ rad}$ $\theta = \dfrac{27.5}{2\pi} = 4.34\text{ rev}$	
6 (b)(i)	 	I A 3 1 2 2 1

b)(ii)	$\sum F_y = 0$ $N_F - W_b - W_P = 0$ $N_F = W_b + W_P$ $N_F = (10)(9.81) + (20)(9.81)$ $N_F = 294.3\ N$ Taking moment about point F (pivot point) $\sum \tau = 0$ $W_b(x) - W_P(2) = 0$ $(10)(9.81)(x) = (20)(9.81)(2)$ $x = 4\ m$	**G1** **JU1** **G1** **JU1**
	TOTAL MARKS	**12**
a)(i)	$\frac{Q}{t} = -\frac{kA(\Delta T)}{L}$ $\frac{Q}{t} = -\frac{kA(T_C - T_H)}{L}$ $Q = -t\frac{kA(T_C - T_H)}{L}$ $= -\frac{(60)(205)(\pi)(0.015)^2(273.15-373.15)}{0.6}$ $= 1449.06\ J$	**G1** **JU1**
a)(ii)	$\frac{Q}{t} = -\frac{kA(\Delta T)}{L}$ $\Delta T = \left(\frac{Q}{t}\right)\left(\frac{L}{kA}\right)$ $= \frac{(96x10^4)(5x10^{-2})}{(96)(100)}$ $= 5\ K$	**G1** **JU1**

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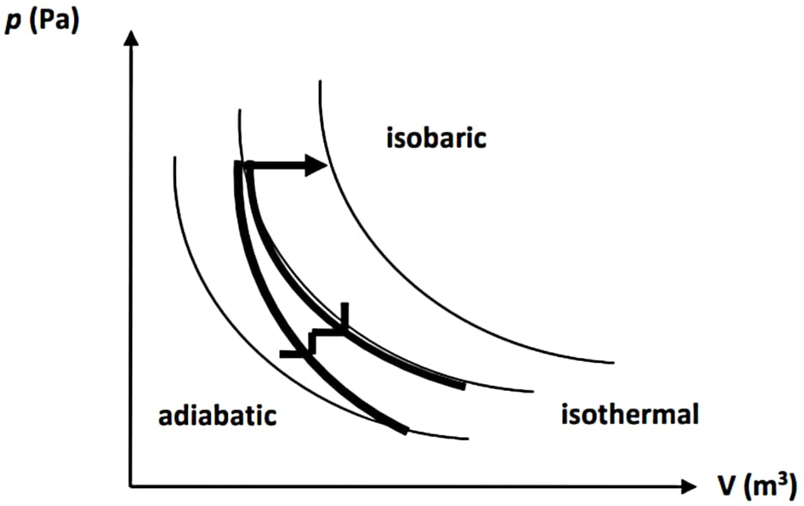
7 (b)(i)	$n = \frac{PV}{RT}$ $= \frac{(153\times10^{3})(515\times10^{-6})}{8.31(322)}$ $= 2.945\times10^{-2}\ moles$ $M = \frac{m}{n}$ $= \frac{0.460}{2.945\times10^{-2}}$ $= 15.6\ g/mol$	
(b)(ii)	$v_{rms} = \sqrt{\frac{3RT}{M}}$ $T = \frac{v_{rms}{}^{2}M}{3R} = \frac{(339)^{2}(0.0440)}{3(8.31)}$ $= 202.83\ \text{K}$	
(b)(iii)	$v_{rms} = \sqrt{\frac{3kT}{m}}$ $= \sqrt{\frac{(3)(1.38x10^{-23})(300)}{\left(\frac{32x10^{-3}}{6.02x10^{23}}\right)}}$ $= 483.4\ \text{ms}^{-1}$	
7 (c)(i)	First Law of Thermodynamics, $Q = \Delta U + W$ $\Delta U = Q - W$ $= 100 - 40$ $= 60\ \text{J}$ Hence, the internal energy is increased by 60 J	

(c)(ii)	
(c)(iii)	$\Delta U = Q - W$ $\Delta U = 100 - (p\Delta V)$ $\Delta U = 100 - (1\times10^{5})\left((500-300)10^{-6}\right)$ $\Delta U = 80J$
	TOTAL MARKS